Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Liver

Date: 22 May 2025
Time: 12:24:09 PM

Female

Gene Symbols	Entrez Gene IDs	Probe IDs	BMD_{rd25} ($BMDL_{rd25}$ - $BMDU_{rd25}$) in ppm	Maximum Fold Change	Direction of Expression Change
Cyp2b2	361523	CYP2B2_32473	48.4 (25.0 - 78.3)	87.1	UP
Cyp2b1	24300	CYP2B1_32451,CYP2B2_32473	69.9 (43.9 - 101.5)	195.0	UP
Abcc3	140668	ABCC3_7941	103.8 (60.2 - 157.7)	4.8	UP
Ugt2b1	286954	UGT2B10_33303,UGT2B1_33224	105.4 (75.3 - 172.0)	5.5	UP
Ephx1	25315	EPHX1_8567	115.8 (61.5 - 183.9)	5.6	UP
Cyp2c6	293989	CYP2C6V1_33169	117.1 (65.4 - 120.1)	3.7	UP
Loc100911718	100911718	CYP2C6V1_33169	117.1 (65.4 - 120.1)	3.7	UP
Car3	54232	CAR3_8196	117.8 (41.2 - 266.8)	2.3	DOWN
Egr1	24330	EGR1_8533	127.0 (30.5 - 318.2)	2.5	DOWN
Aldh1a1	24188	ALDH1A1_8022	127.6 (110.1 - 130.6)	8.0	UP

rd25 = 25% relative deviation

Annotations:

Descriptions of orthologous human genes are shown due to the increased detail that is available in public resources such as UniprotKB (http://www.uniprot.org/uniprot/) and Entrez Gene (https://www.ncbi.nlm.nih.gov/gene/). Human UniprotKB was used as the primary resource due to the greater breadth of annotation and depth of functional detail that is provided. Rat UniprotKB was used as the second resource if the primary source did not provide a detailed description of function. Human Entrez gene summary was used as third resource. Rat Entrez gene summary was used as the fourth resource. **Gene definition version:** https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

Cyp2b2: Human Uniprot function (Human CYP2B6): A cytochrome P450 monooxygenase involved in the metabolism of endocannabinoids and steroids (PubMed:21289075, PubMed:12865317). Mechanistically, uses molecular oxygen inserting one oxygen atom into a substrate, and reducing the second into a water molecule, with two electrons provided by NADPH via cytochrome P450 reductase (NADPH–hemoprotein reductase) Catalyzes the epoxidation of double bonds of arachidonoylethanolamide (anandamide) to 8,9-, 11,12-, and 14,15-epoxyeicosatrienoic acid ethanolamides (EpETrE-EAs), potentially modulating endocannabinoid system signaling (PubMed:21289075). Hydroxylates steroid hormones, including testosterone at C-16 and estrogens at C-2 (PubMed:21289075, PubMed:12865317). Plays a role in the oxidative metabolism of xenobiotics, including plant lipids and drugs (PubMed:11695850, PubMed:22909231). Acts as a 1,4-cineole 2-exo-monooxygenase (PubMed:11695850). \{ECO:0000269;PubMed:11695850, ECO:0000269; PubMed:12865317, ECO:0000269;PubMed:21289075, ECO:0000269;PubMed:22909231\}.; FUNCTION: Allele 2B6*9: Has low affinity for anandamide and can only produce 11,12 EpETrE-EAs. \{ECO:0000269; PubMed:21289075\}.

Cyp2b1: Human Uniprot function (Human CYP2B6): A cytochrome P450 monooxygenase involved in the metabolism of endocannabinoids and steroids (PubMed:21289075, PubMed:12865317). Mechanistically, uses molecular oxygen inserting one oxygen atom into a substrate, and reducing the second into a water molecule, with two electrons provided by NADPH via cytochrome P450 reductase (NADPH–hemoprotein reductase). Catalyzes the epoxidation of double bonds of arachidonoylethanolamide (anandamide) to 8,9-, 11,12-, and 14,15-epoxyeicosatrienoic acid ethanolamides (EpETrE-EAs), potentially modulating endocannabinoid system signaling (PubMed:21289075). Hydroxylates steroid hormones, including testosterone at C-16 and estrogens at C-2 (PubMed:21289075, PubMed:12865317). Plays a role in the oxidative metabolism of xenobiotics, including plant lipids and drugs (PubMed:11695850, PubMed:22909231). Acts as a 1,4-cineole 2-exo-monooxygenase (PubMed:11695850). \{ECO:0000269;PubMed:11695850, ECO:0000269;PubMed:12865317, ECO:0000269;PubMed:21289075, ECO:0000269;PubMed:22909231\}.; FUNCTION: Allele 2B6*9: Has low affinity for anandamide and can only produce 11,12 EpETrE-EAs. \{ECO:0000269;PubMed:21289075\}.

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Abcc3: Human Uniprot function (Human ABCC3): May act as an inducible transporter in the biliary and intestinal excretion of organic anions. Acts as an alternative route for the export of bile acids and glucuronides from cholestatic hepatocytes (By similarity). \{ECO:0000250\}.

Ugt2b1: Human Uniprot function (Human UGT2B17): UDP-glucuronosyltransferase (UGT) that catalyzes phase II biotransformation reactions in which lipophilic substrates are conjugated with glucuronic acid to increase the metabolite's water solubility, thereby facilitating excretion into either the urine or bile (PubMed:8798464, PubMed:16595710, PubMed:18719240, PubMed:19022937, PubMed:23288867). Catalyzes the glucuronidation of endogenous steroid hormones such as androgens (epitestosterone, androsterone) and estrogens (estradiol, epiestradiol) (PubMed:8798464, PubMed:16595710, PubMed:18719240, PubMed:19022937, PubMed:23288867). {ECO:0000269|PubMed:16595710, ECO:0000269|PubMed:18719240, ECO:0000269|PubMed:19022937, ECO:0000269|PubMed:23288867, ECO:0000269|PubMed:8798464}.

Ephx1: Human Uniprot function(Human EPHX1): Biotransformation enzyme that catalyzes the hydrolysis of arene and aliphatic epoxides to less reactive and more water soluble dihydrodiols by the trans addition of water (By similarity). Plays a role in the metabolism of endogenous lipids such as epoxide-containing fatty acids (PubMed:22798687). Metabolizes the abundant endocannabinoid 2-arachidonoylglycerol (2-AG) to free arachidonic acid (AA) and glycerol (PubMed:24958911). \{ECO:0000250;UniProtKB:P07687, ECO:0000269;PubMed:22798687,ECO:0000269;PubMed:24958911\}.

Cyp2c6: Rat Uniprot function (Rat Cyp2c6): Cytochromes P450 are a group of heme-thiolate monooxygenases. In liver microsomes, this enzyme is involved in an NADPH-dependent electron transport pathway. It oxidizes a variety of structurally unrelated compounds, including steroids, fatty acids, and xenobiotics.

Loc100911718: No description available

Car3: Rat Uniprot function (Rat Car3): Reversible hydration of carbon dioxide

Egr1: Human Uniprot function (Human EGR1): Transcriptional regulator (PubMed:20121949). Recognizes and binds to the DNA sequence 5'-GCG(T/G)GGGCG-3'(EGR-site) in the promoter region of target genes (By similarity). Binds double-stranded target DNA, irrespective of the cytosine methylation status (PubMed:25258363, PubMed:25999311). Regulates the transcription of numerous target genes, and thereby plays an important role in regulating the response to growth factors, DNA damage, and ischemia. Plays a role in the regulation of cell survival, proliferation and cell death. Activates expression of p53/TP53 and TGFB1, and thereby helps prevent tumor formation. Required for normal progress through mitosis and normal proliferation of hepatocytes after partial hepatectomy. Mediates responses to ischemia and hypoxia; regulates the expression of proteins such as IL1B and CXCL2 that are involved in inflammatory processes and development of tissue damage after ischemia. Regulates biosynthesis of luteinizing hormone (LHB) in the pituitary (By similarity). Regulates the amplitude of the expression rhythms of clock genes: ARNTL/BMAL1, PER2 and NR1D1 in the liver via the activation of PER1 (clock repressor) transcription. Regulates the rhythmic expression of core-clock gene ARNTL/BMAL1 in the suprachiasmatic nucleus (SCN) (By similarity). {ECO:0000250|UniProtKB:P08046, ECO:0000269|PubMed:20121949, ECO:0000269|PubMed:25258363, ECO:0000269|PubMed:25999311}.

Aldh1a1: Human Uniprot function (Human Aldh1a1): Can convert/oxidize retinaldehyde to retinoic acid. Binds free retinal and cellular retinol-binding protein-bound retinal (By similarity). May have a broader specificity and oxidize other aldehydes in vivo (PubMed:19296407, PubMed:26373694, PubMed:25450233).

** END OF REPORT **